

SSH

What is Linux

- Not straightforward
- Technically a **kernel**
 - Core of the operating system

What is Linux?

- Many operating systems use the Linux kernel
 - Ubuntu (EOS)
 - Arch, Kali, Gentoo, Fedora, Manjaro, Mint, etc.
- These OS's are often referred to as “Linux”

Unix

- Earlier operating system
- Linux and other systems are heavily influenced by Unix
- We now use “Unix-like” systems”
 - Linux
 - macOS
 - FreeBSD
 - Solaris
 - Etc.

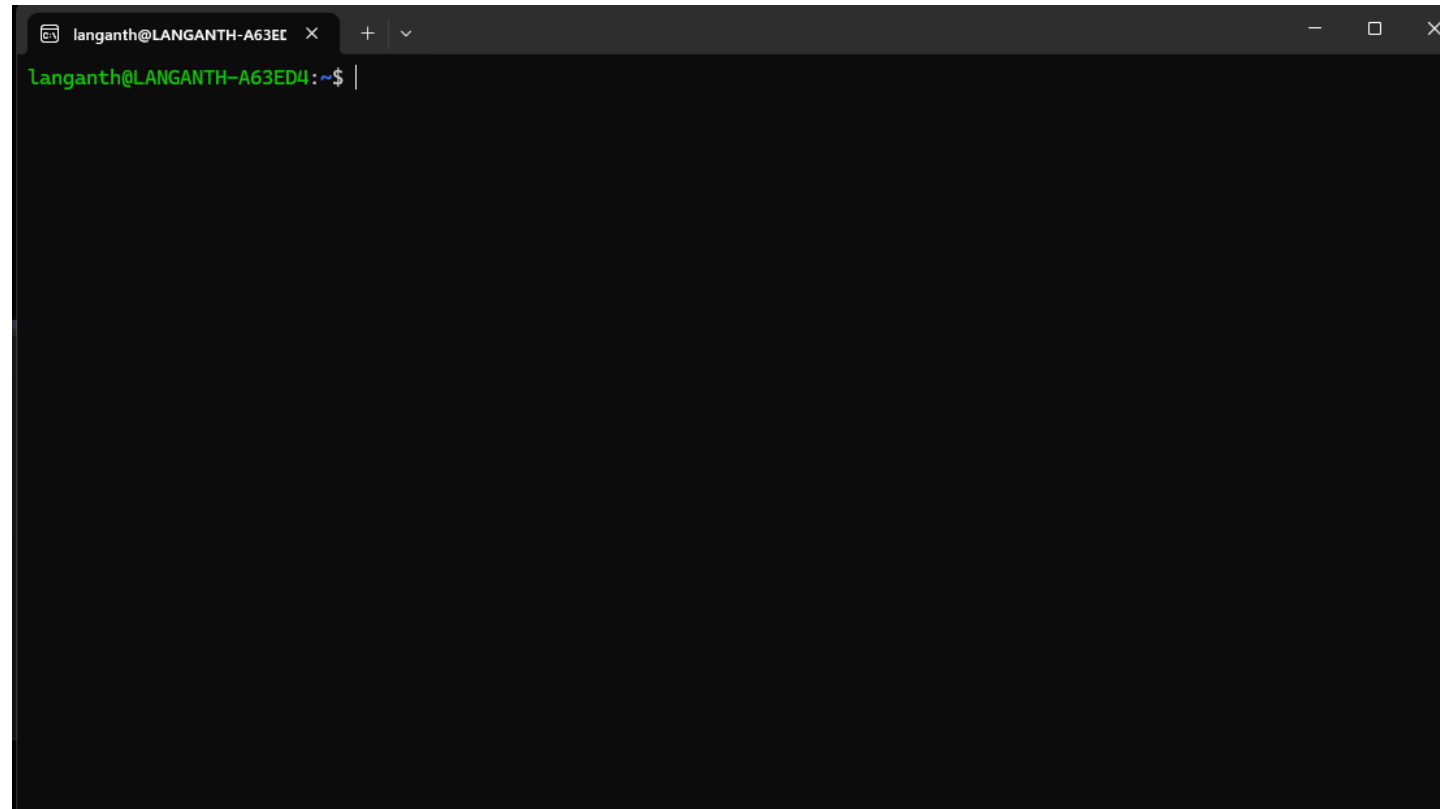
Windows

- Windows is not Unix-like
- Can you use Windows for this class?
 - By default? No!
 - **W**indows **S**ubsystem for **L**inux (WSL)
 - If you are on Windows, you **must** install WSL

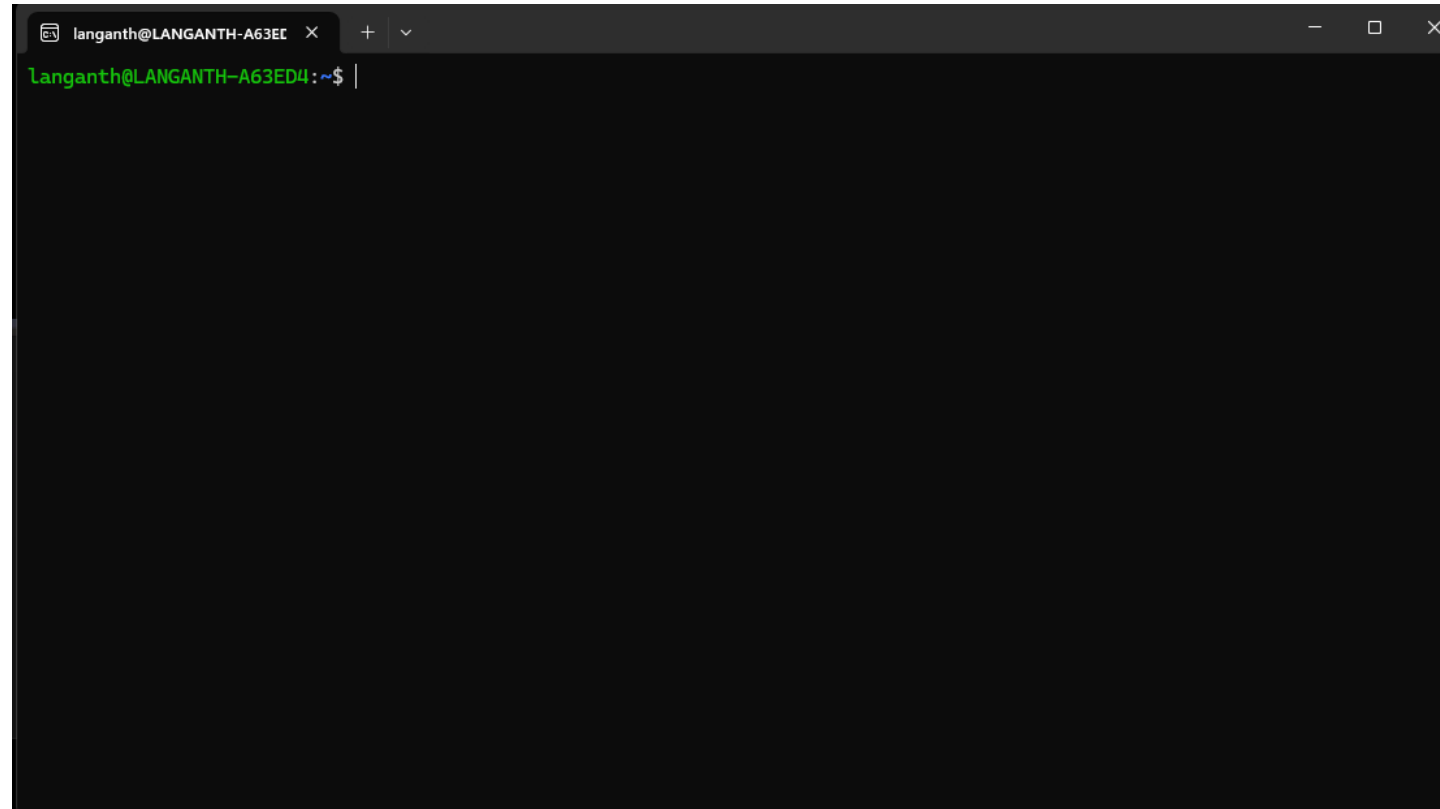
Shell

- Terminal
- We are running a shell
- What is a shell?
 - A shell is a way to pass through commands to the operating system
 - Give command → Computer go brrrrr

What is this?

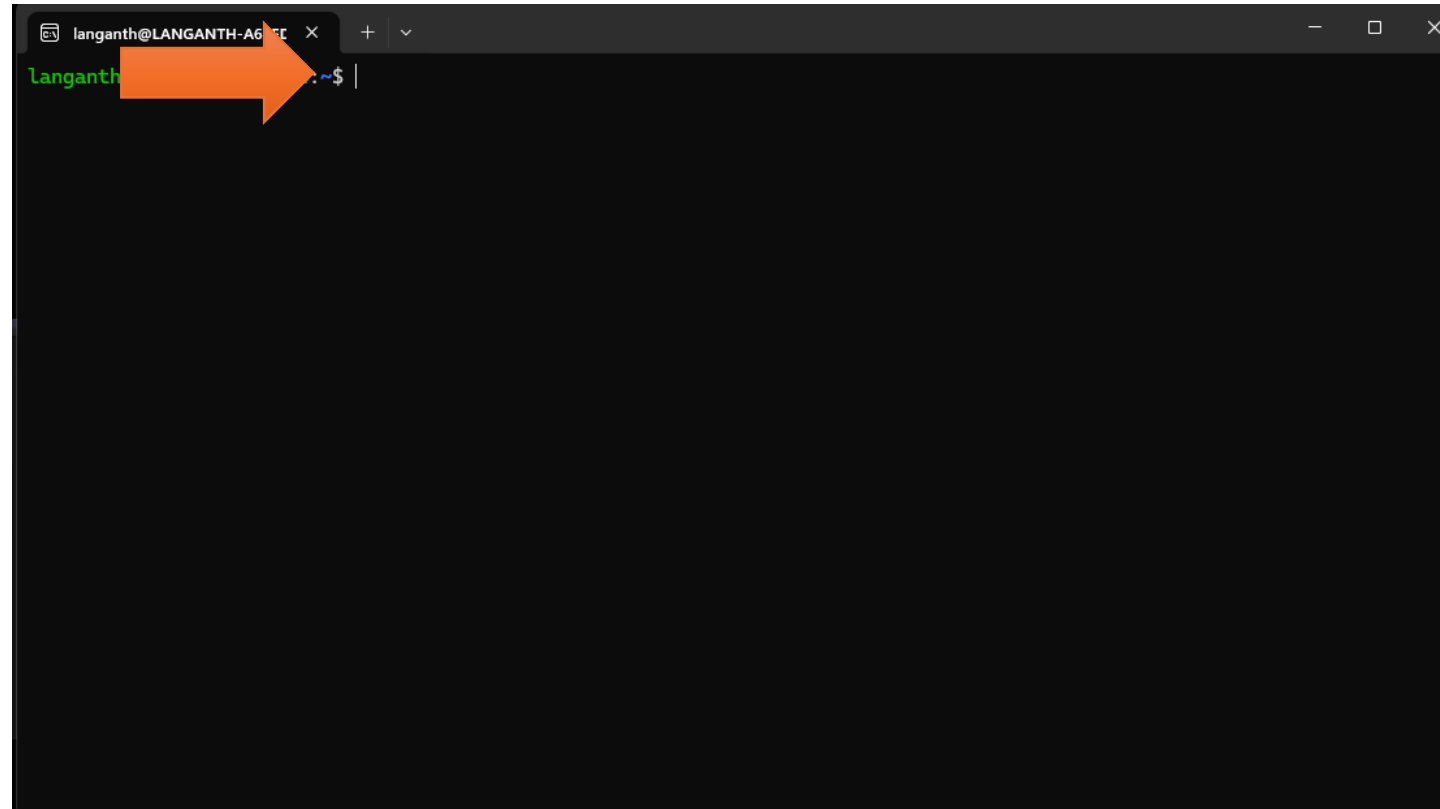


What is this?

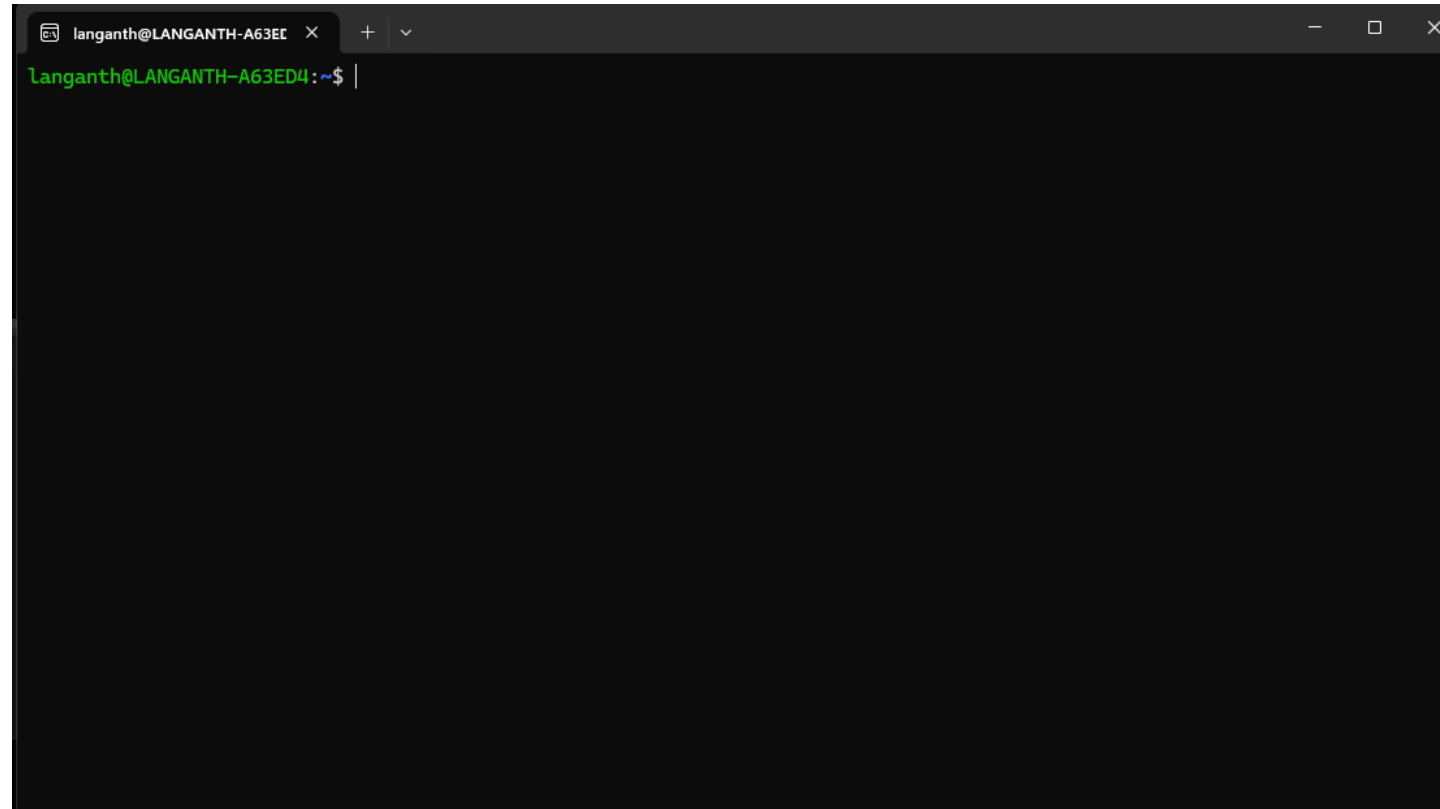


- A terminal!

What about this?



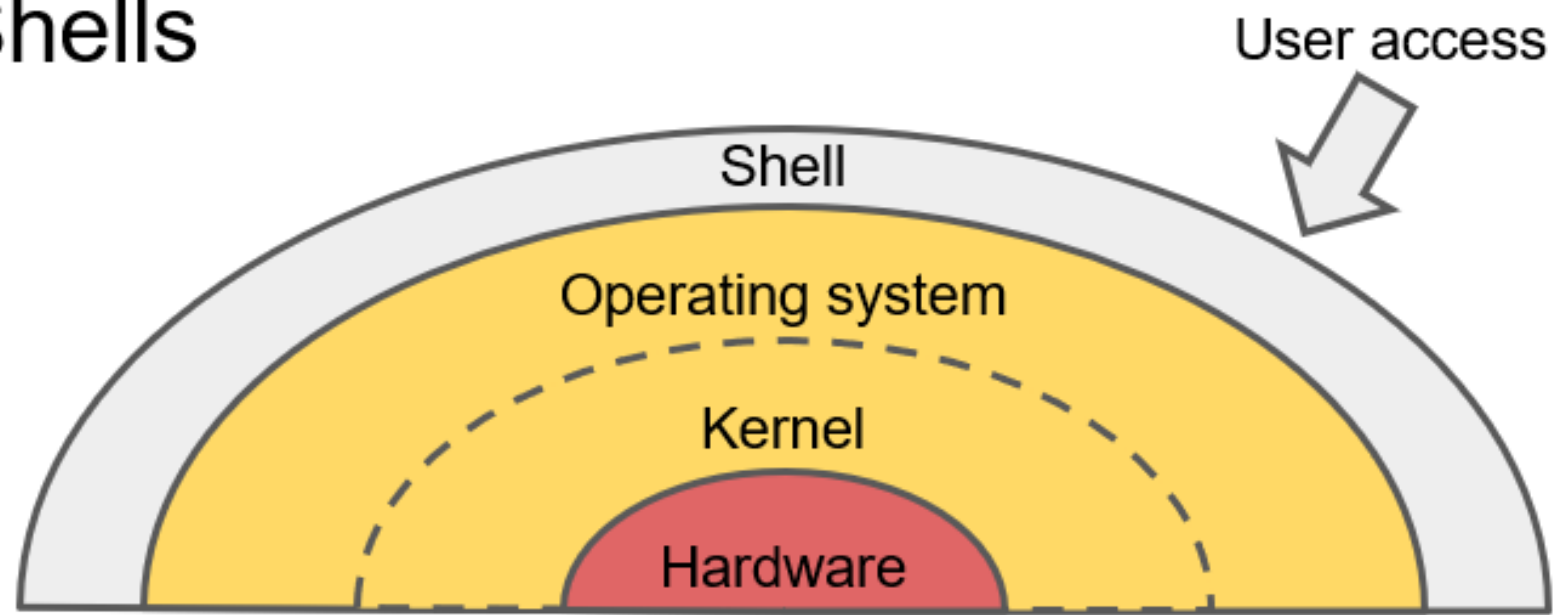
What about this?

A screenshot of a terminal window. The window has a dark gray title bar with a single tab labeled 'langanth@LANGANTH-A63EE'. Below the title bar, the terminal content shows a green prompt 'langanth@LANGANTH-A63ED4:~\$' followed by a white cursor. The rest of the terminal area is black and empty.

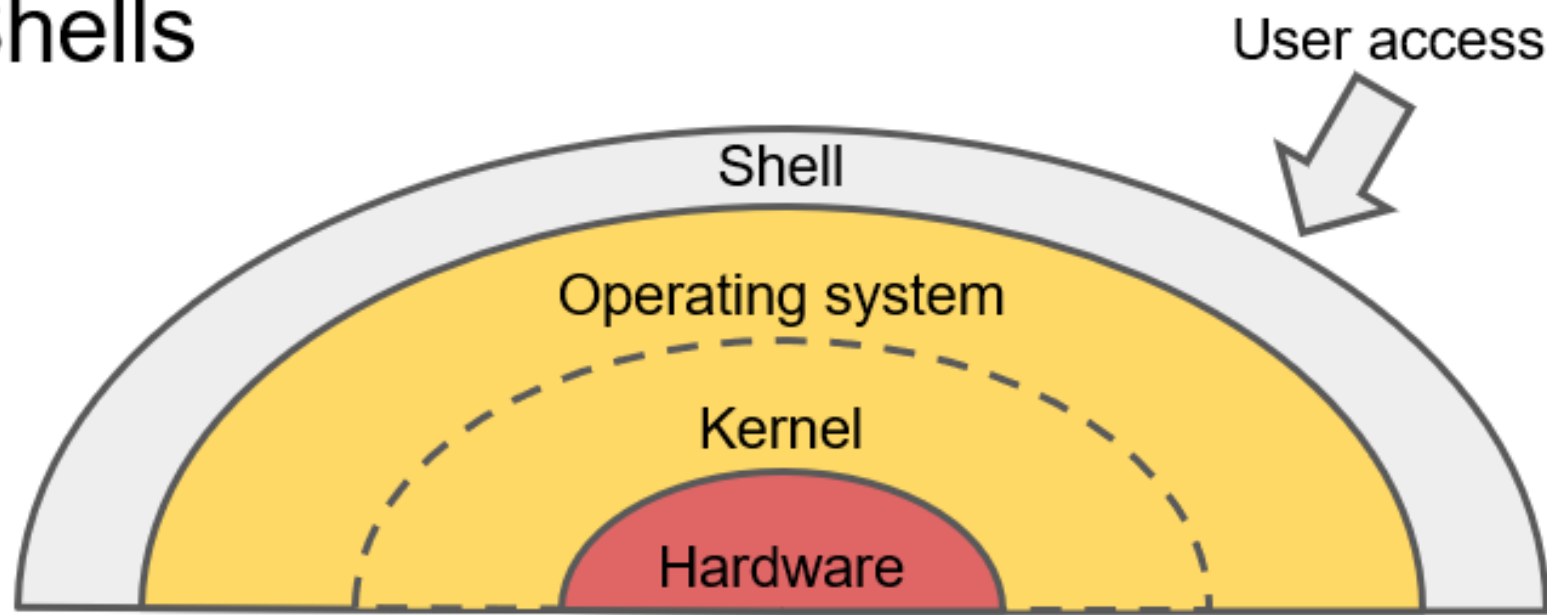
```
langanth@LANGANTH-A63EE x + v - □ x  
langanth@LANGANTH-A63ED4:~$ |
```

- We are running a **shell**

Shells



Shells



```
langanth@LANGANTH-A63EE  X  +  v
langanth@LANGANTH-A63ED4:~$ echo $0
-bash
langanth@LANGANTH-A63ED4:~$ |
```

What is echo?

- Think print
- \$0?
 - Command being run

Connecting to another machine?

- ssh – **S**ecure **S**hell
 - Ssh allows us to securely connect to a remote shell
 - How to use?
 - ssh username@address

langanth@eos01: ~

langanth@LANGANTH-A63ED4:~\$ echo \$0

-bash

langanth@LANGANTH-A63ED4:~\$ ssh langanth@eos01.cis.gvsu.edu

The authenticity of host 'eos01.cis.gvsu.edu (35.39.29.66)' can't be established.

ED25519 key fingerprint is SHA256:YWT5v4IwJf6amH2McgLYoegBbRedFkbuRB1DSundxIo.

This key is not known by any other names.

Are you sure you want to continue connecting (yes/no/[fingerprint])? yes

Warning: Permanently added 'eos01.cis.gvsu.edu' (ED25519) to the list of known hosts.

langanth@eos01.cis.gvsu.edu's password:

Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 7.0.0-29-generic x86_64)

System information as of Mon Aug 31 11:42:18 AM EDT 2026

System load:	0.09	Temperature:	126.0 C
Usage of /:	65.3% of 97.87GB	Processes:	325
Memory usage:	19%	Users logged in:	0
Swap usage:	0%	IPv4 address for enp1s0:	35.39.29.66

EOS Lab

Contact HPC Support for help/questions

E-Mail : arcit@gvsu.edu

langanth@eos01:~\$ |

Connecting to another machine?

- Once you do this, you will be running commands on another computer!
- But why?
 - Efficiency
 - Necessity
 - Only have shell access
 - Tools only available from command line

Wait! How do I leave?!

- exit